



Food and Agriculture
Organization of the
United Nations

وزارة البيئة والمياه والزراعة
Ministry of Environment Water & Agriculture



Guideline on organic beekeeping practices



Guideline on organic beekeeping practices

*Nuru Adgaba Mohammed
International Consultant*

Required citation:

Mohammed, N.A, 2026. *Guideline on organic beekeeping practices*. Riyadh, FAO. <https://doi.org/10.4060/cd8167en>

The designations employed and the presentation of material in this information product do not imply the expression of any opinion whatsoever on the part of the Food and Agriculture Organization of the United Nations (FAO) concerning the legal or development status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. The mention of specific companies or products of manufacturers, whether or not these have been patented, does not imply that these have been endorsed or recommended by FAO in preference to others of a similar nature that are not mentioned.

The views expressed in this information product are those of the author(s) and do not necessarily reflect the views or policies of FAO.

ISBN 978-92-5-140426-3

© FAO, 2026



Some rights reserved. This work is made available under the Creative Commons Attribution- 4.0 International licence (CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/legalcode.en>).

Under the terms of this licence, this work may be copied, redistributed and adapted, provided that the work is appropriately cited. In any use of this work, there should be no suggestion that FAO endorses any specific organization, products or services. The use of the FAO logo is not permitted. If a translation or adaptation of this work is created, it must include the following disclaimer along with the required citation: "This translation [or adaptation] was not created by the Food and Agriculture Organization of the United Nations (FAO). FAO is not responsible for the content or accuracy of this translation [or adaptation]. The original [Language] edition shall be the authoritative edition."

Any dispute arising under this licence that cannot be settled amicably shall be referred to arbitration in accordance with the Arbitration Rules of the United Nations Commission on International Trade Law (UNCITRAL). The parties shall be bound by any arbitration award rendered as a result of such arbitration as the final adjudication of such a dispute.

Third-party materials. This Creative Commons licence CC BY 4.0 does not apply to non-FAO copyright materials included in this publication. Users wishing to reuse material from this work that is attributed to a third party, such as tables, figures or images, are responsible for determining whether permission is needed for that reuse and for obtaining permission from the copyright holder. The risk of claims resulting from infringement of any third-party-owned component in the work rests solely with the user.

FAO photographs. FAO photographs that may appear in this work are not subject to the above-mentioned Creative Commons licence. Queries for the use of any FAO photographs should be submitted to: photo-library@fao.org.

Sales, rights and licensing. FAO information products are available on the FAO website (www.fao.org/publications) and print copies can be purchased through the distributors listed there. For general enquiries about FAO publications please contact: publications@fao.org. Queries regarding rights and licensing of publications should be submitted to: copyright@fao.org.

Cover photograph: © Nuru Adgaba Mohammed

Contents

Foreword.....	vi
Acknowledgements.....	vii
Abbreviations	viii
1. Introduction.....	1
2. Definition and principles of organic production.....	2
2.1 Definition of organic production.....	2
2.2 Principles of organic production	2
3. General requirements for organic beekeeping	3
3.1 Definition of organic honey	5
4. Procedures to be followed in starting and conversion into organic beekeeping.....	6
4.1 Selection of suitable apiary sites	6
4.2 Source and origin of bees, beehives, and inputs.....	7
5. Operational requirements for organic beekeeping	7
5.1 Feeding practices.....	8
5.2 Bee health management.....	9
5.2.1 Approved treatments for honeybee diseases and pests	9
5.3 Colony and apiary management	10
5.4 Migratory organic beekeeping	10
5.5 Record keeping and traceability.....	11
6. Procedures for pre and postharvest handling of organic honey	12
6.1 Harvesting and processing.....	12
6.2 Storage and transport	12
7. Conversion period	14
8. Organic certification	15
9. Conclusion	15
10. References	16

Figures

1. Flowchart depicting the major steps of organic beekeeping and its certification.....	4
2. Example of isolated organic apiary site with natural and diverse vegetations.....	6
3. Organic migratory beekeeping.....	11
4. Organic honeys kept in food grade glass jars	13



Foreword

The Food and Agriculture Organization of the United Nations (FAO) provides technical support to the Ministry of Environment, Water and Agriculture (MEWA) and Saudi REEF in implementing the project “*Strengthening MEWA's capacity to implement its Sustainable Rural Agricultural Development (SRAD) Programme.*” This strategic initiative aims to enhance institutional capacities, boost agricultural productivity, strengthen rural institutions, generate employment, and empower youth- and women-led enterprises — all while promoting the sustainable management of land, water, and other vital natural resources.

Today, due to increasing awareness of human health issues and growing environmental concerns for environment, organic products have become popular and are steadily increasing in demand. Organic production is an integrated approach covering the entire value chain from production to consumption. It emphasizes natural production methods, avoiding chemicals, genetically modified organisms, hormones, antibiotics, and artificial inputs. Organic beekeeping practices are also expanding globally, with over three million bee hives certified for organic beekeeping worldwide. Therefore, having comprehensive and dedicated organic beekeeping practice guidelines—covering all aspects from production to post-harvest handling and marketing of bee products—is essential.

The current guideline is prepared mainly in the context of beekeeping and includes more topics which were not covered well in the other guidelines. It is developed mainly by adopting the Codex Alimentarius guidelines (1999) for the production, processing, labeling and marketing of organically produced foods and the International Federation of Organic Agriculture Movement (IFOAM) definitions and principles of organic production. The guideline consists of general definition and principles of organic production, selection of suitable apiary sites for organic beekeeping, procedures for establishment and operation of organic beekeeping, management bee colonies and apiary under organic beekeeping standards, maintaining honeybee health and disease control in organic beekeeping, migratory organic beekeeping practices, procedures for pre- and postharvest handling of organic honey and the steps of organic certification.



Dr. Nizar Haddad
FAO Saudi Arabia Programme Director



Acknowledgements

The Food and Agriculture Organization of the United Nations wishes to express its sincere appreciation to all those who contributed to the preparation of the publication “Guideline on Organic Beekeeping Practices.”

The author extends his sincere gratitude to the FAO Saudi Arabia team and the senior members of the FAO-Sustainable Rural Agricultural Development (SRAD) Programme, particularly Nizar Haddad, Programme Director, and Kakoli Ghosh, Chief Technical Advisor, for their continuous guidance and steadfast support throughout the development of this guideline.

Special appreciation is also extended to the Department of Organic Production at the Ministry of Environment, Water and Agriculture (MEWA), Saudi Arabia, and to Raed Saleh Almusaylim for his ongoing support during the implementation of organic beekeeping and the pilot demonstration of organic beekeeping practices in the Kingdom of Saudi Arabia.

The author acknowledges Reyhan Bahtiyarca valuable review, which significantly enhanced the quality of this publication; to Mafaz Punjabi for her excellent design work on the document; and to Zunita Zubir for her invaluable assistance and guidance in formatting and final editing in preparation for publication.



Abbreviations

FAO	Food and Agriculture Organization of the United Nations
SRAD	Sustainable Rural Agricultural Development
MEWA	Ministry of Environment, Water and Agriculture
IFOAM	International Federation of Organic Agriculture Movement



1. Introduction

Today due to the increasing concern for our environment and increasing knowledge and public awareness on human health issues, organic products become more popular. As a result of high demand and increased production, organic products have become mainstream and one of the rapidly growing sectors in food industry. Organic production is an integrated approach that includes the whole value chain from the producers to the consumers and focuses on production of products naturally avoiding chemicals, genetically modified products, hormones, antibiotics, and any artificial inputs.

There is also a shift in the beekeeping world, from conventional beekeeping practices to organic beekeeping practices. In 2018 about 2.6 million beehives all over the world have been registered as organic (Pocol, *et al.*, 2021) and it is steadily increasing, and the number reaches over three million by 2019 (Bioeco Actual, 2021).

Since honey is a cash crop that is mainly produced for market purposes, it is very important to build consumer's trust in the quality of honey and the way it is produced. Building local honey consumers' trust, through introducing organic and ethical beekeeping practices is important pillar for the sustainability of beekeeping. Moreover, organic honey has many relative advantages over conventional honey such as increased marketability, easy traceability (due to the mandatory labelling and recording system), product quality assurance, and better price incentives. Moreover, once certification for organic beekeeping is obtained, bee products other than honey (beeswax, bee collected pollen, propolis, royal jelly and others) can also be sold as organic products without requiring additional registration and certifications process.

Moreover, organic beekeeping practice contributes to positive attitudinal changes of beekeepers motivating them to follow good beekeeping practices and encourages integrative approaches with positive effects on the health of humans, honeybees, and environment that in turn contributes to long-term sustainability of beekeeping sector.

Since organic beekeeping commonly started through conversion of the existing conventional beekeeping practices, the aim of this practical guideline is to assist beekeepers in the process of converting to organic beekeeping and to demonstrate specific procedures and aspects of organic beekeeping. The document only covers aspects of organic beekeeping activities those differ from the conventional practices.



2. Definition and principles of organic production

2.1 Definition of organic production

Codex Alimentarius Commission, of the United Nations (1999) described organic agriculture as “holistic production management system which promotes and enhances agro-ecosystem health, including biodiversity, biological cycles, and soil biological activity” It encourages the use of locally adapted systems, cultural, biological, and mechanical methods and opposes the use of synthetic materials. According to the description, an organic production system is designed to enhance biological diversity, soil fertility, recycling of waste, relying on renewable resources and minimizing all forms of pollution that ensure organic honey production.

According to the International Federation of Organic Agriculture Movement (IFOAM) General Assembly (2008), Organic Agriculture is defined as: “a production system that sustains the health of soils, ecosystems, and people. Organic agriculture relies on ecological processes, biodiversity and cycles adapted to local conditions, rather than the use of inputs with adverse effects. Organic Agriculture combines tradition, innovation, and science to benefit the shared environment and promote fair relationships and good quality of life for all involved”.

Generally, a beekeeping to be considered as organic should fulfill all the requirements organic beekeeping practices which mainly include using of locally adapted bee race, keeping at suitable foraging area, use of organic inputs, stay for sufficient conversion periods, mandatory inspections and certification by authorized body.

2.2 Principles of organic production

IFOAM (2024), focuses on four major principles in organic production, ecology, health, fairness and care. The principle of ecology is to attain ecological balance, that focuses on conserving the whole ecosystem, saving the environment from pollution, enhance biological diversity, relying on renewable resources, recycling, and reusing inputs for better quality environment, encourage the healthy use of soil, water, and air.

From a beekeeping point of view, ecology is very important because it relates to an environment where beekeepers keep their bees for better honey production. If ecology is not suitable for beekeeping, skill, knowledge, and technologies by themselves do not make beekeeping successful. So, beekeepers in general and organic practitioners in particular should play a great role in maintaining ecological balances.



In relation to principles of health, organic farming focuses on production of products naturally, which is based on protecting the health of human beings that encourages minimizing or totally avoiding of use of chemicals/pesticides, genetically modified products, hormones, antibiotics and any artificial inputs that may lead to negative health effects.

In the principle of fairness, organic production supports equity, respect and justice (benefit sharing) fairly to all groups (producers, processors, distributors, suppliers and consumers). In the principle of care: organic farming is required to manage in a precaution and ethical manner to save the well-being of present and future generations and the environment.

3. General requirements for organic beekeeping

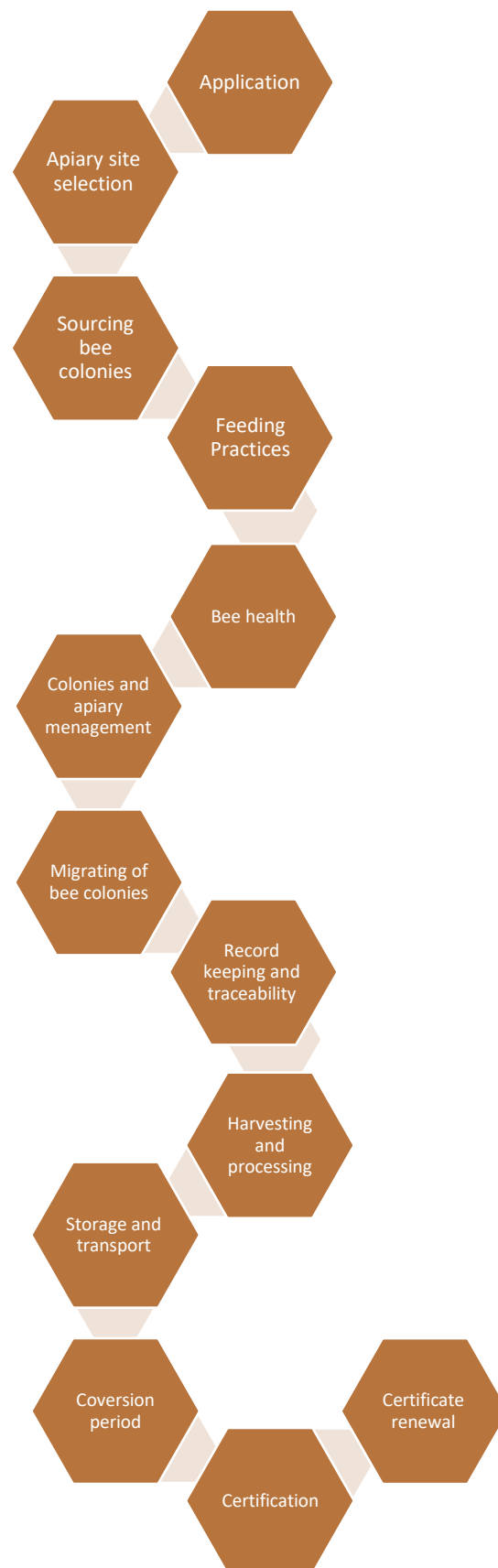
The general practices of organic beekeeping are the same as that of other organic production. However, the practices for organic beekeeping and honey production slightly differ from those for other organic livestock products.

Further, organic beekeeping differs from the conventional beekeeping in that, it needs to follow the specific procedures at different levels of honey production and processing to ensure that the steps follow match with the procedures of organic honey production.

Some of the major steps/actions required for organic beekeeping practices and certifications are shown below in the form of a flowchart as Figure 1.



Figure 1. Flowchart depicting the major steps of organic beekeeping and its certification



Source: Author's own elaboration.

3.1 Definition of organic honey

Organic honey is a honey that has been produced naturally according to specific standards, avoiding chemical or biological contamination with non-organic materials at every stage of its production and processing. Organic beekeeping is not only whether the honey being organic or not, but it includes the principles or ethics that protect and enhance the health of the entire ecosystem during the process of honey production.



4. Procedures to be followed in starting and conversion into organic beekeeping

4.1 Selection of suitable apiary sites

- Bee colonies for organic beekeeping should be placed in areas where the nectar and pollen sources plant are natural wild vegetation and herbs or if cultivated crops, they should be organically produced;
- the selected apiary site should be at least 3 km away from other non-organic beekeeping apiary sites and away from any chemical and industrial contaminant sources;
- bee colonies should be kept away from potential contaminants and forbidden substances, genetically modified plants and organisms;
- the respective organic certification official body or authority must approve the selected areas to ensure the appropriateness of the site and the nectar and pollen sources for organic beekeeping; and
- the official certification body may specify the radius of the apiary site where the bees must forage.

Figure 2. Example of isolated organic apiary site with natural and diverse vegetations



© Nuru Adgaba Mohammed



4.2 Source and origin of bees, beehives, and inputs

Origin of bees: Bees and the queens used or purchased for organic beekeeping must come from certified organic beekeeping apiaries if available. If this is not possible, the bee colonies must undergo a conversion period (one year) before organic certification can be claimed. In the selection of bee colonies, the ability of bees to cope-up with the local environmental conditions, and their ability to withstand certain honeybee diseases should be considered. In relation to this, for organic beekeeping indigenous bees should be favored over exotic/imported bees. Queen for replacement should come from the same or other organic apiaries.

Beehives: Beehives used for organic beekeeping should be made from natural materials (non-chemical treated timber), but it is permitted to use metal sheets for the external hive covers. Each hive should have its own hive identity number label which is important for record keeping and traceability.

Beeswax: For modern (box) hives, the new beeswax foundation sheets should be sourced from organic certified production units. In traditional beehives, the naturally secreted beeswax can be used as organic beeswax if the bees are kept under organic beekeeping conditions.



5. Operational requirements for organic beekeeping

5.1 Feeding practices

- Colonies with organic beekeeping, during honey harvesting, beekeeper must leave with adequate amount of honey and pollen to support the bees in the dearth period without artificial feeding;
- bee colonies under organic beekeeping should naturally survive without artificial feeding, however, under exceptional circumstances like disease outbreak, occurring of temporary feed shortages due to climatic conditions such as elongated drought or other, supplementary feeding of colonies can be allowed to withstand the problem;
- in such situations, if available honey or sugar from organic source can be used;
if pollen or pollen substitutes or supplements are prepared and feed it must come from organic production;
- however, under certain circumstances, the certification authority can allow the feeding of non-organically produced honey or sugars. But the feeding time should be limited, and the organic production should be temporarily held;
- in such cases feeding should be carried out at least one month before the commencing of the next nectar flow;
- the sugar provided should not exceed the requirements of the colony and should not be stored in the comb not to mix with natural nectar or honey; and
- if any excess stored sugar remains in the hive, it must be totally removed before the start of nectar flow.



5.2 Bee health management

In organic beekeeping the honeybee colonies' health must be managed through applying good beekeeping practices, emphasizing prevention measures which include:

- use indigenous bees well adapted to the local conditions;
- use disease resistant bee breeds;
- regular cleaning and disinfecting of equipment with approved organic disinfectants (Lactic acid, Oxalic acid, Acetic acid, Formic acid and others) or use of boiling water or flame to disinfect hives, frames and tools;
- regular renewal of beeswax, (in each hive one third of the old combs should be replaced every year);
- assuring the presence of adequate resources (pollen and honey) to maintain the bee colonies in dearth period;
- diseased bee colonies must be moved to isolated areas (about 3 km) away from healthy colonies;
- contaminated beehives and materials that cannot be decontaminated by the above-described cleaning and sterilizing procedures need to be destroyed;
- apply only natural products to treat honeybee diseases (like thymol, oxalic acid and formic acid and others);
- avoid veterinary medicines (antibiotics) and acaricides; and
- apply biomechanical means to minimize the effects of some honeybee diseases such as:
 - ❖ for varroa disease: by interrupting the mites life cycle through minimizing or creating absence of honeybee brood in a colony either by removing the broods or temporarily confining the honeybee queen to limit her egg laying and subsequently not to have a brood period so that the varroa mites cannot complete their life cycle to reinfect the colony; and
 - ❖ For chalkbrood, remove all infected combs and broods and replace them with new foundation sheets.

5.2.1 Approved treatments for honeybee diseases and pests

- Organic products recommended by Codex Alimentarius (1999) to treat some honeybee diseases and pests in organic beekeeping include: lactic, oxalic, acetic, and formic acids can be used;
- natural essential oils such as menthol, eucalyptol, and camphor can be applied;
- the treatments should be strictly used according to the manufacturer's guidelines given on the labeling of the products;
- sterilization of infested hives and equipment with steam and direct flame, where natural preventative methods are not working, registered and approved veterinary medicinal products can be applied, but;
- priority should be given first to herbal or plant origin and alternative homemade treatments;



- if chemically produced products are used, the certifying body should be informed, and the harvested honey and other bee products should not be sold as organic products;
- after chemically synthesized medicinal treatment, the beeswax must be replaced completely with organic beeswax;
- every veterinary treatment used must be clearly documented and reported to certifying body; and
- the treated bee colonies should be kept isolated and should undergo another conversion period of one year.

5.3 Colony and apiary management

- The bee colonies should be optimally managed to remain healthy and strong through accessing sufficient sources of pollen and nectar;
- the bee colonies should access sufficient and uncontaminated fresh water;
- the bee colonies should be protected from direct sunlight and heat either by keeping under natural trees shade or man-made shelters or covering the hive with sunlight reflecting materials;
- during honey harvesting the killing of bees on the combs is not allowed in organic beekeeping;
- in organic beekeeping clipping of the wings of honeybee queens not allowed;
- during honey harvesting the use of non-natural repellent substance is prohibited;
- smoking should be moderate, and materials used for smoking should be natural plant materials;
- during harvesting sufficient honey should be left to the colony to maintain itself up to the next nectar flow period;
- apiary should be kept clean;
- do not discard plastics and waste around apiary and it should be managed following the principles of ecology;
- beeswax must not be discarded in apiary and should be collected, melted and recycled; and
- reuse of inputs by recycling waste for better quality of environment and optimum utilization of available resources is one of the important principles of organic production.

5.4 Migratory organic beekeeping

Organic beekeeping through migration of bee colonies is possible as far as the selected sites where colonies are moving comply with the requirements for organic beekeeping. However, when colonies are moved, the organic certification body should be informed, to inspect the new site and management of colonies as required.



If necessary, apiary tracking devices Global Positioning System (GPS) can be used to track the movement of colonies by organic certifiers.

Colonies to being migrated should be packed to:

- secure the position various hive components (hive cover and frames) placed firmly;
- during transportation avoid shaking frames and hives;
- provide adequate ventilation to the bees;
- prevent congestion inside the hive;
- prevent honeybees escaping through gaps in entrance gates, and other hive components; and
- move the bee colonies during the night and at moderate temperature.

In migratory organic beekeeping, colonies should be kept at least 3 km away from non-organic beekeeping colonies, which is to avoid the bees accessing non-organic bee feeds used by non-organic beekeepers.

Figure 3. Organic migratory beekeeping



© FAO/ Fayeze Muhammed Al-Dhubyani

5.5 Record keeping and traceability

- Each colony in organic beekeeping must have its own hive reference number;
- total number of colonies and honey yield of each colony and total honey production should be recorded and reported;
- organic beekeepers should keep detailed and latest records about their organic beekeeping practices; and
- the maps where all colonies are placed should be shared with the certifying body.



6. Procedures for pre and postharvest handling of organic honey

6.1 Harvesting and processing

- During honey harvesting protective clothing and personal hygiene should be maintained;
- during harvesting the smoking material should not be synthetic and meet the organic beekeeping requirements;
- bee repellents consisting of prohibited substances (chemical synthetic) must not be used;
- excessive smoke shall not be used as it may taint the flavor of honey or otherwise spoil its aroma and tastes;
- harvest only ripe honey without open and sealed brood;
- the place used for honey extraction shall be hygienic and free from insects (flies, ants and bees);
- during the extraction and processing of honey, the temperatures should be as low as possible (less than 40 °C);
- honey extractors and filters must be food graded and suitable for organic beekeeping;
- honey extractors and filters must be cleaned without chemical detergent or soap; and
- the honey should be kept in food grade containers

6.2 Storage and transport

- Organic honey must not be mixed with non-organic honey and products;
- honey shall be stored in food graded and air-tight containers;
- the honey shall be stored in a cold room (preferably ≤ 20 °C) and away from direct sunlight;
- organic honey must be kept away from substances not allowed to use for organic production;
- stores for organic honey must be separated from non-organic honey and products and must be labeled clearly;
- in case organic and non-organic honeys are kept in the same store there must be proper labeling;
- containers for transporting organic honey must be washed with materials allowed for organic production; and
- care must be taken to avoid any contamination of organic honey with pesticide or other contaminants.



Figure 4. Organic honeys kept in food grade glass jars



© Nuru Adgaba Mohammed



© Nuru Adgaba Mohammed

7. Conversion period

Honey and hive products can be labelled and sold as organic products, if the organic production procedures presented here have been followed for a period of a minimum of one year. During the conversion time all the previously used beeswax in the target hives should be changed with organic beeswax. If no prohibited substances have been used before in the hive, the beeswax from honey uncapping and from traditional hives can be used upon getting authorization from certifying bodies. If the guidelines are not complied, the conversion period is going to be extended.



8. Organic certification

Organic Certification: is a process by which formally authorized/mandated certification organizations, issue a proof of letter ensuring that the product, process, or services conforms with certain organic standards. Any product to be labeled and sold as an organic product must be certified by such an officially authorized body. Even if the beekeeping practices comply with the requirements of organic production, beekeepers cannot claim by themselves that the product is “organic” by default unless the authorized body inspect and certifies the honey and its production process as “organic”. Once the beekeeping practice is certified, as organic beekeeping, all the hive bee products and by products (honey, beeswax, pollen, propolis, royal jelly and bee colonies) obtained from the same apiary can be labeled and sold as organic products.

Anyone who is interested in getting the certification for organic beekeeping must apply either to the authorized certifying body or to the local regulatory body for organic production. Once the beekeeper has applied, the application will be reviewed by the concerned expert, from an authorized organic certifying body. Then the authorized organic certifying body will make the necessary regular inspection which depends on migration of colonies and major honey harvesting seasons and auditing the apiary and the beekeeping practices. Detailed procedures and requirements for organic certification will be provided to the applicant by the certifying body. The fees for regular inspection and certification are expected to be covered by the applicant. Organic certification is mostly valid for only one year and must be renewed every year fulfilling the required procedures for organic beekeeping.

9. Conclusion

The guidelines will be used by beekeepers, development and extension agents to introduce and implement organic beekeeping practices. The information besides serving as guidelines for organic beekeeping practices will be useful for both organic and non-organic beekeepers to implement integrated, sustainable and ethical beekeeping practices caring for the health of bees, humans and the environment.



10. References

Bio Eco Actual. 2021. Organic honey: premium and on the rise. *International Organic Newspaper*. [Cited 11 August 2025]. <https://www.bioecoactual.com/en/2021/06/09/>.

Codex Alimentarius. GL 32-1999. 1999. Guidelines for the production, processing, labelling and marketing of organically produced foods. (Adopted 1999, Revisions 2001, 2003, 2004 and 2007; Amendments 2008 and 2009). Codex Alimentarius. Rome. https://www.ipcinfo.org/fileadmin/user_upload/livestockgov/documents/cxg_032e.pdf.

Pocol, C.B., Šedík, P., Bruměa, I.S., Amuza, A. & Chirsanova, A. 2021. Organic Beekeeping Practices in Romania: Status and Perspectives towards a Sustainable Development. *Agriculture*, 11(4): 281. <https://doi.org/10.3390/agriculture11040281>.

International Federation of Organic Agriculture Movements. 2008. Definition of organic agriculture. Bonn. IFOAM. <https://www.ifoam.bio/why-organic/organic-landmarks/definition-organic>.

International Federation of Organic Agriculture Movements. Four principles of organic agriculture. Bonn, IFOAM. <https://www.ifoam.bio/why-organic/shaping-agriculture/four-principles-organic>.

FAO. 2001. Appendix II. Guideline for the production, processing, labelling and marketing of organically produced foods (livestock and livestock products): Beekeeping and additives. [Accessed September 2024]. <http://www.fao.org/4/y0651e/y0651e0l.htm>.





FAO Technical Cooperation Programme in Saudi Arabia

**Strengthening MEWA's capacity to implement its Sustainable Rural
Agricultural Development Programme (SRAD)**

(UTF/SAU/051/SAU)

With the financial support of:



FAO Technical Cooperation Programme in Saudi Arabia
Email: Fao-sa@fao.org

Food and Agriculture Organization of the United Nations
Riyadh, Saudi Arabia

